

Report for setting up a raspberry pi kit

CE6461 – Computer Vision Systems

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Introduction:

This report outlines the setup and verification of a computer vision platform consisting of a Raspberry Pi 4 and outlines the steps to move it to a networked remote access over Ethernet. The work preconditions the further laboratories with vision because it prepares the RPi4 kit that includes a built-in camera and proves the build with the help of the photo documentation that includes the Student ID card, the setting up of the network parameters, and the implementation of the command-line access via SSH.

The kit is first assembled as per the instructions given, and all the procedures involved in connecting the camera module and related peripherals are handled carefully. The next steps are configuring and test connectivity to the network by means of obtaining the Raspberry Pi Ethernet and Wi-Fi MAC addresses via the console that identifies the device in the laboratory network. Finally, the direct wired connection is found to be effective because it is possible to open an SSH session to the Raspberry Pi via Ethernet, and a terminal screenshot is registered as the empirical evidence.

These steps altogether ensure that the hardware is properly assembled, and the device is uniquely identifiable on the network, and the secure remote access is provided. The resulting artefacts, i.e. a photograph of the put-together kit with the Student ID card, a console screenshot of the MAC addresses demarcation, and a terminal screenshot of a SSH login are verifiable checkpoints that show that the platform is ready to be experimented with computer vision.

Building a Raspberry Pi 4 kit



Figure: Assembled Raspberry Pi 4 kit

Console window showing your Ethernet and WiFi MAC addresses

```
File Edit Tabs Help
sharon@raspberrypi:~$ ipconfig
bash: ipconfig: command not found
sharon@raspberrypi:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.137.10 netmask 255.255.255.0 broadcast 192.168.137.255
    inet6 fe80::8df8:fea:28f8:c103 prefixlen 64 scopeid 0x20<link>
    ether d8:3a:dd:af:6a:18 txqueuelen 1000 (Ethernet)
    RX packets 2392 bytes 164191 (160.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8563 bytes 11276620 (10.7 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 50 bytes 5395 (5.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 50 bytes 5395 (5.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.178.126 netmask 255.255.255.0 broadcast 192.168.178.255
    inet6 2a02:278:2018:c1:32b4:8ec5:cc7b:77f7 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::2a62:b70d:7061:b456 prefixlen 64 scopeid 0x20<link>
    inet6 fd00:69fd:8bdc:0:c2e:72ff:d8c6:2c6c prefixlen 64 scopeid 0x0<global>
    ether d8:3a:dd:af:6a:19 txqueuelen 1000 (Ethernet)
    RX packets 426 bytes 105918 (103.4 KiB)
    RX errors 0 dropped 107 overruns 0 frame 0
    TX packets 161 bytes 21077 (20.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

sharon@raspberrypi:~$
```

SSH login to RPi4 over direct Ethernet

```
s25249347@raspberrypi: ~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\shall> ssh s25249347@192.168.137.10
s25249347@192.168.137.10's password:
Linux raspberrypi 6.12.25+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.25-1+rpt1 (2025-04-30) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Sep 20 14:04:56 2025
s25249347@raspberrypi:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether d8:3a:dd:af:6a:18 brd ff:ff:ff:ff:ff:ff
    inet 192.168.137.10/24 brd 192.168.137.255 scope global noprefixroute eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::8df8:fea:28f8:c103/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether d8:3a:dd:af:6a:19 brd ff:ff:ff:ff:ff:ff
    inet 192.168.178.126/24 brd 192.168.178.255 scope global dynamic noprefixroute wlan0
        valid_lft 862112sec preferred_lft 862112sec
    inet6 2a02:278:2018:c1:32b4:8ec5:cc7b:77f7/64 scope global dynamic noprefixroute
        valid_lft 6785sec preferred_lft 3185sec
    inet6 fd0:69fd:8bdc:0:c2e:72ff:d8c6:2c6c/64 scope global dynamic noprefixroute
        valid_lft 6785sec preferred_lft 3185sec
    inet6 fe80::2a62:b70d:7061:b456/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
s25249347@raspberrypi:~$
```

Conclusion

This lab has created an operational Raspberry 4 computer-vision system by following systematic methods of assembling the kit- including identification of network identity through Ethernet and Wi-Fi MAC address and confirmation of direct wired connectivity by successfully logging into SSH. The artefacts required, which are a photo of the completed kit with the Student ID near it and the terminal screenshots are the key elements of proofs of the accomplishment of every specified milestone. Therefore, once hardware setting, network connectivity, and remote access requirements are satisfied positively, the system will be ready to execute the next image acquisition and processing processes in the future laboratories.